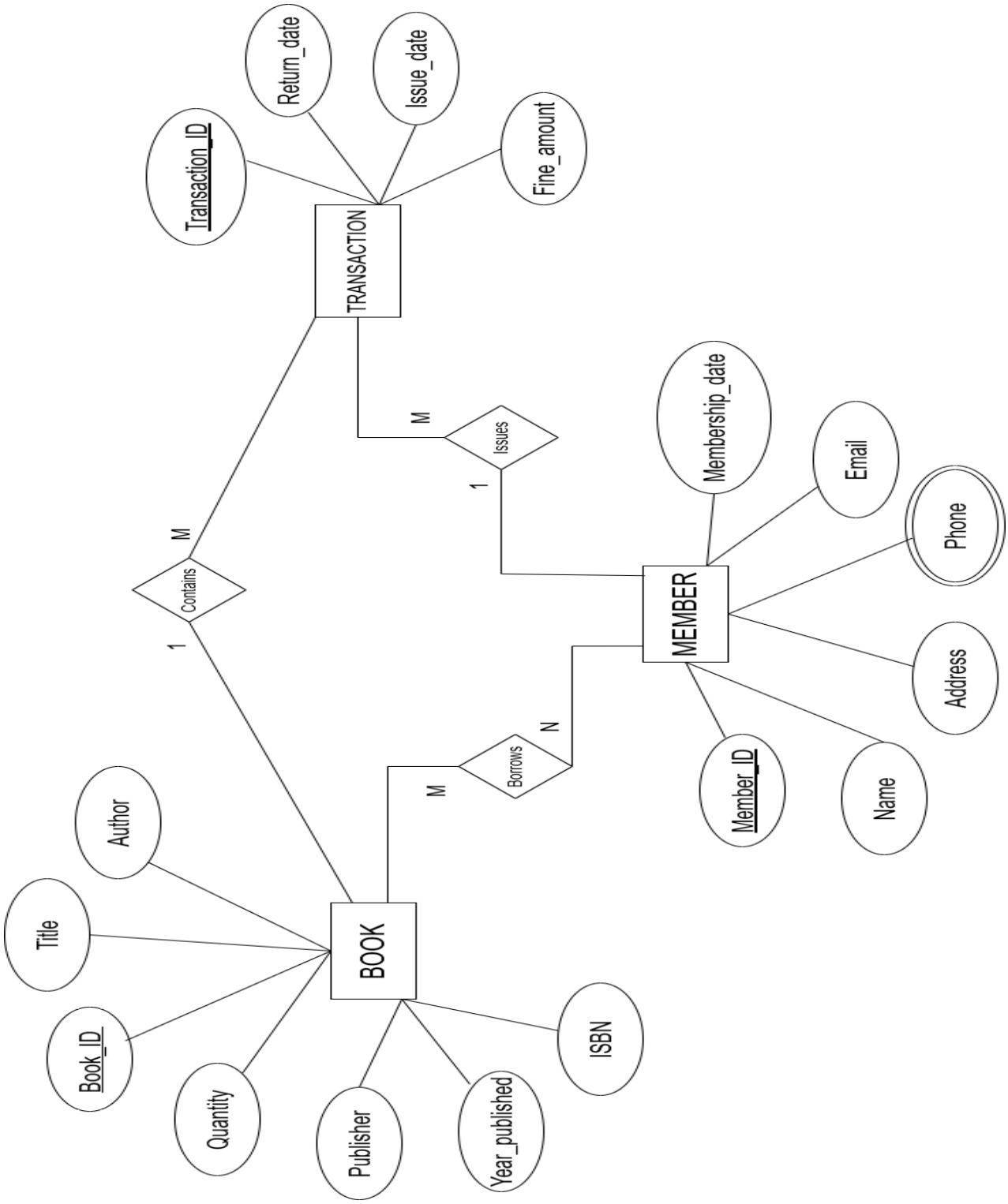
EXP – 1:


Transaction_ID
Return_date
Issue_date
Fine_amount
TRANSACTION
M
Contains
1
M
Issues
1
Membership_date
Email
Phone
MEMBER
Address
Name
Member_ID
N
Borrows
M
Author
Title
BOOK
Book_ID
Quantity
Publisher
Year_published
ISBN

EXP – 2:

Initial data:

Output:

student_id	name	dept	year	email
1	Rahul	AI	2023	rahul@gmail.com
2	Neha	ML	2024	neha@gmail.com
3	Arun	DS	2025	arun@gmail.com

After ALTERING:

Output:

student_id	name	dept	email	phone
1	Rahul	AI	rahul@gmail.com	NULL
2	Neha	ML	neha@gmail.com	NULL
3	Arun	DS	arun@gmail.com	NULL

EXP – 3:

Initial data:

Output

emp_id	emp_name	salary	age	email	dept_id
1	Rahul	50000	26	rahul@gmail.com	HR
2	Neha	60000	18	neha@gmail.com	Finance
6	Nina	60000	25	nina@gmail.com	IT
3	Arun	55000	19	arun@gmail.com	Marketing

NOT NULL Constraint:

Employee

emp_id	emp_name	salary	email
1	Rahul	50000	26
2	Neha	60000	18
6	Nina	60000	25
3	Arun	55000	17

CHECK Constraint:

emp_id	emp_name	salary	age
1	Rahul	50000	26
2	Neha	60000	18
6	Nina	60000	25

PRIMARY KEY and UNIQUE Constraint:

Employee

emp_id	emp_name	salary	age	email
1	Rahul	50000	26	rahul@gmail.com
2	Neha	60000	18	neha@gmail.com
6	Nina	60000	25	nina@gmail.com
3	Arun	55000	19	arun@gmail.com

FOREIGN KEY Constraint:

Employee

emp_id	emp_name	salary	age	email
1	Rahul	50000	26	rahul@gmail.com
2	Neha	60000	18	neha@gmail.com
6	Nina	60000	25	nina@gmail.com
3	Arun	55000	19	arun@gmail.com

Department

dept_id	dept_name
901	HR
802	Marketing
907	IT
705	Finance

Output

emp_id	emp_name	salary	age	email	dept_id
1	Rahul	50000	26	rahul@gmail.com	HR
2	Neha	60000	18	neha@gmail.com	Finance
6	Nina	60000	25	nina@gmail.com	IT
3	Arun	55000	19	arun@gmail.com	Marketing

EXP – 4:

Initial data:

Output

stud_id	st_name	branch	CGPA	company	package	status
305	Anya	CSE	8.88	Infosys	8.68	Employed
235	Arjun	AIML	7.88	Infosys	9.68	Employed
347	Akash	Mechanical	9.8	TATA MOTORS	4.68	Employed
364	Aboli	Robotics	6.48	WIPRO	6.68	Employed
330	Arsh	CSE	9	MICROSOFT	8.66	Employed
399	Bheem	CSE	3.08		0	Unemployed
498	Anuj	Civil	7.88	ESRI	5.78	Employed
389	Duryodhan	Robotics	6.98	GOOGLE	6.8	Employed

Show only employed students:

Output

stud_id	st_name	branch	CGPA	company	package	status
305	Anya	CSE	8.88	Infosys	8.68	Employed
235	Arjun	AIML	7.88	Infosys	9.68	Employed
347	Akash	Mechanical	9.8	TATA MOTORS	4.68	Employed
364	Aboli	Robotics	6.48	WIPRO	6.68	Employed
330	Arsh	CSE	9	MICROSOFT	8.66	Employed
389	Duryodhan	Robotics	6.98	GOOGLE	6.8	Employed

Show students with package greater than 8 LPA:

Output

stud_id	st_name	branch	CGPA	company	package_lpa	status
305	Anya	CSE	8.88	Infosys	8.68	Employed
235	Arjun	AIML	7.88	Infosys	9.68	Employed
330	Arsh	CSE	9	MICROSOFT	8.66	Employed

Show CSE students with CGPA greater than 8:

Output

stud_id	st_name	branch	CGPA	company	package_lpa	status
305	Anya	CSE	8.88	Infosys	8.68	Employed
330	Arsh	CSE	9	MICROSOFT	8.66	Employed

Display names and packages sorted by highest package:

Output

stud_id	st_name	branch	CGPA	company	package_lpa	status
235	Arjun	AIML	7.88	Infosys	9.68	Employed
305	Anya	CSE	8.88	Infosys	8.68	Employed
330	Arsh	CSE	9	MICROSOFT	8.66	Employed
389	Duryodhan	Robotics	6.98	GOOGLE	6.8	Employed
364	Aboli	Robotics	6.48	WIPRO	6.68	Employed
498	Anuj	Civil	7.88	ESRI	5.78	Employed
347	Akash	Mechanical	9.8	TATA MOTORS	4.68	Employed
399	Bheem	CSE	3.08		0	Unemployed

Count how many students are placed:

Output

COUNT(status)
6

Increase package by 1 LPA for all Infosys company students:

Output

stud_id	st_name	branch	CGPA	company	package_lpa	status
305	Anya	CSE	8.88	Infosys	9.68	Employed
235	Arjun	AIML	7.88	Infosys	10.68	Employed

EXP – 5:

Initial data:

Output:

emp_id	name	salary	bonus	dept	join_date
1	Amit	50000.00	5000.00	IT	2022-01-10
2	Neha	62000.00	7000.00	HR	2021-03-15
3	Ravi	45000.00	4000.00	IT	2023-07-01
4	Sara	70000.00	9000.00	Finance	2020-11-20
5	Karan	52000.00	6000.00	IT	2022-09-05

NUMBER FUNCTIONS:

1. ABS –

   Output:

name	diff_bonus
Amit	3000.00
Neha	1000.00
Ravi	4000.00
Sara	1000.00
Karan	2000.00

2. ROUND –

   Output:

name	divided_salary
Amit	16666.67
Neha	20666.67
Ravi	15000.00
Sara	23333.33
Karan	17333.33

3. CEIL – Next Highest Integer –

Output:

```
+-------------+
| CEIL(45.12) |
+-------------+
|          46 |
+-------------+
```

4. POWER –

Output:

```
+------------+
| POWER(2,5) |
+------------+
|         32 |
+------------+
```

5. MOD –

Output:

```
+-----------+
| MOD(17,5) |
+-----------+
|         2 |
+-----------+
```

AGGREGATE FUNCTIONS:

1. COUNT –

Output:

```
+----------------+
| total_employees |
+----------------+
|              5 |
+----------------+
```

2. SUM –

Output:

```
+--------------+
| total_salary |
+--------------+
|    279000.00 |
+--------------+
```

3. AVG –

Output:

```
+--------------+
| avg_salary   |
+--------------+
| 55800.000000 |
+--------------+
```

4. MAX –

Output:

```
+---------------+
| highest_salary |
+---------------+
|      70000.00 |
+---------------+
```

5. MIN –

Output:

```
+---------------+
| lowest_salary |
+---------------+
|      45000.00 |
+---------------+
```

CONVERSION FUNCTIONS:

1. CAST(value AS datatype) –

   Output:

```
+---------------------+
| CAST(salary AS CHAR) |
+---------------------+
| 50000.00            |
| 62000.00            |
| 45000.00            |
| 70000.00            |
| 52000.00            |
+---------------------+
```

2. CAST(string AS SIGNED) –

   Output:

```
+------------------------+
| CAST('1234' AS SIGNED) |
+------------------------+
|                   1234 |
+------------------------+
```

3. CONVERT(value, datatype) –

   Output:

```
+----------------------------+
| CONVERT('2025-01-01', DATE) |
+----------------------------+
| 2025-01-01                 |
+----------------------------+
```

4. CAST(number AS DECIMAL(p,s)) –

   Output:

```
+--------------------------------+
| CAST(123.4567 AS DECIMAL(10,2)) |
+--------------------------------+
|                         123.46 |
+--------------------------------+
```

DATE FUNCTIONS:

1. Current Date –

   Output:

```
+------------+
| CURDATE()  |
+------------+
| 2026-02-23 |
+------------+
```

2. Current Timestamp –

   Output:

```
+---------------------+
| NOW()               |
+---------------------+
| 2026-02-23 16:30:12 |
+---------------------+
```

3. Year Extraction –

   Output:

```
+-------+-----------------+
| name  | YEAR(join_date) |
+-------+-----------------+
| Amit  |            2022 |
| Neha  |            2021 |
| Ravi  |            2023 |
| Sara  |            2020 |
| Karan |            2022 |
+-------+-----------------+
```

4. Month Extraction –

   Output:

```
+-------+------------------+
| name  | MONTH(join_date) |
+-------+------------------+
| Amit  |                1 |
| Neha  |                3 |
| Ravi  |                7 |
| Sara  |               11 |
| Karan |                9 |
+-------+------------------+
```

5. Date Difference –

   Output:

```
+--------------+
| days_worked  |
+--------------+
|         1505 |
|         1806 |
|          968 |
|         1921 |
|         1267 |
+--------------+
```

EXP – 6:

Initial data:

Output

emp_id	name_emp	dept	salary	city
1	Aryan	IT	50000	Delhi
2	Neha	HR	62000	Mumbai
3	Ravi	IT	45000	Delhi
4	Sara	Finance	70000	Pune
5	Karan	IT	52000	Mumbai
6	Meena	HR	48000	Delhi

GROUP BY clause:

Output

dept	AVG(salary)
Finance	70000
HR	55000
IT	49000

HAVING clause:

Output

dept	AVG(salary)
Finance	70000
HR	55000

ORDER BY clause:

Output

name_emp	salary
Sara	70000
Neha	62000
Karan	52000
Aryan	50000
Meena	48000
Ravi	45000

INDEXING:

Creating Index -

Output:

Table	Non_unique	Key_name	Seq_in_index	Column_name	Collation	Cardinality	Sub_part	Packed	Null	Index_type	Comment	Index_comment	Visible	Expression
employee	0	PRIMARY	1	emp_id	A	2	NULL	NULL		BTREE			YES	NULL
employee	1	indx_dept	1	dept	A	3	NULL	NULL	YES	BTREE			YES	NULL

Dropping Index:

Table	Non_unique	Key_name	Seq_in_index	Column_name	Collation	Cardinality	Sub_part	Packed	Null	Index_type	Comment	Index_comment	Visible	Expression
employee	0	PRIMARY	1	emp_id	A	2	NULL	NULL		BTREE			YES	NULL